PATTERN RECOGNITION AND MACHINE LEARNING

Robust Low-Rank Matrix Recovery Under Sparse Corruptions

A Comparative Study of PCA and Principal Component Pursuit

[Presenter Name]

Motivation and hypothesis

Recovery problem

- Shared structure across samples or features
- Sparse, high-amplitude errors in individual entries
- Example: persistent video background and localized foreground motion [3]

Study question

- Recovery of the clean matrix and the corruption locations
- Comparison with a known-rank SVD baseline on identical inputs

H1

- Lower reconstruction error for PCP under incoherent low-rank structure and random sparse errors, with a wider advantage at higher corruption density

Synthetic data and experiment parameters

$$\underbrace{M}_{\text{observed}} = \underbrace{L_0}_{\text{clean signal}} + \underbrace{S_0}_{\text{sparse errors}}, \qquad L_0 = \frac{AB^\top}{\sqrt{r}}, \quad A_{ik},\, B_{jk} \overset{\text{iid}}{\sim} \mathcal{N}(0,1).$$

Signal and corruption

- Matrix size 100×100, true rank $r = 5$
- Unit entry variance in L_0 from $1/\sqrt{r}$ scaling
- round(pmn) distinct locations, sampled uniformly
- Independent signs, magnitude 10 at each selected entry

Trial design

- Density $p \in \{0, 2, 5, 10, 20\}\%$
- 5 independent draws per density, 25 main trials
- Seed 42 with separate trial streams
- Shared M per trial for both methods

Truncated SVD baseline

$$M = U\Sigma V^\top, \qquad \hat{L}_{\mathrm{PCA}} = U_r \Sigma_r V_r^\top, \qquad \hat{L}_{\mathrm{PCA}} \in \arg\min_{\mathrm{rank}(L)\le r} \|M - L\|_{\mathrm{F}}^2 .$$

- Top $r = 5$ singular directions, with the true rank supplied
- Optimal squared-error approximation to M at rank r [1]
- Recovery evaluated against L_0, a different target from the fitted M
- Uncentered SVD for raw matrix recovery
- Centered PCA on $M - \mathbf{1}\bar{m}^\top$, with the mean restored for reconstruction in the original coordinates [2]

Principal Component Pursuit

$$\min_{L,S} \ \|L\|_* + \lambda \|S\|_1 \quad \text{subject to} \quad L + S = M \qquad \lambda = \frac{1}{\sqrt{\max(m,n)}} = 0.100.$$

Convex penalties

- Nuclear norm $\|L\|_* = \sum_i \sigma_i(L)$, favoring low rank
- Entrywise norm $\|S\|_1 = \sum_{ij} |S_{ij}|$, favoring sparse errors
- Rank and corruption locations inferred jointly

PCP formulation and conditional recovery guarantees [3]

Identifiability

- Incoherent signal directions, spread across coordinates
- Sufficiently low rank and scattered corruption support
- Single-entry matrix as an ambiguous case, both sparse and rank one

Scaled ADMM updates

$$\mathcal{L}_\mu = \|L\|_* + \lambda\|S\|_1 + \frac{\mu}{2}\|M - L - S + Y\|_{\mathrm{F}}^2 - \frac{\mu}{2}\|Y\|_{\mathrm{F}}^2.$$

$$L^{k+1} = \mathrm{SVT}_{1/\mu}\left(M - S^k + Y^k\right),$$
$$S^{k+1} = \mathrm{Soft}_{\lambda/\mu}\left(M - L^{k+1} + Y^k\right),$$
$$Y^{k+1} = Y^k + M - L^{k+1} - S^{k+1}.$$

- L step: shrink singular values via $\sigma \mapsto (\sigma - t)_+$ [4]
- S step: shrink entries via $z \mapsto \mathrm{sgn}(z)(|z| - t)_+$
- Dual step: accumulate the constraint mismatch [5]
- $S^0 = Y^0 = 0$, fixed $\mu = 0.3\sqrt{mn}/\|M\|_{\mathrm{F}}$ for nonzero M
- Cap of 5000 iterations

Evaluation protocol

Signal reconstruction

$$E_L = \frac{\left\| \hat{L} - L_0 \right\|_{\mathrm{F}}}{\|L_0\|_{\mathrm{F}}}.$$

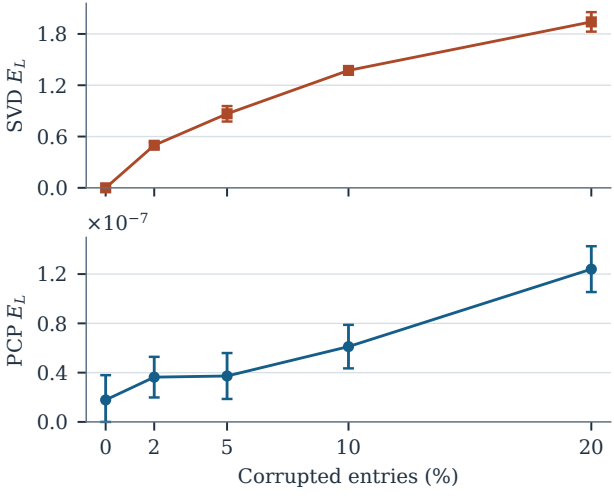
- Mean and sample SD across 5 trials per density
- First 10% trial for the component and residual plots
- Solver time, iteration counts, and convergence status

Corruption detection

$$\hat{\Omega} = \{(i,j) : |\hat{S}_{ij}| > 1\}.$$

- Predicted mask against true support Ω
- Precision, recall, F1, and false-positive rate
- Empty true support at 0%: undefined recall and F1, false alarms reported separately

Reconstruction error across corruption densities



Mean $\pm$ one sample SD, with separate vertical scales

- Clean control: SVD near machine precision
- Lower PCP error in 20/20 corrupted trials
- At 20%: 1.941 for SVD, 1.24×10^{-7} for PCP
- Widening mean error gap across the tested sweep

Why the reconstruction errors diverged

Measured error by density

p	Truncated SVD	PCP
0%	$1.82 \times 10^{-15} \pm 1.80 \times 10^{-15}$	$1.78 \times 10^{-8} \pm 2.01 \times 10^{-8}$
2%	0.498 ± 0.036	$3.63 \times 10^{-8} \pm 1.65 \times 10^{-8}$
5%	0.866 ± 0.090	$3.73 \times 10^{-8} \pm 1.86 \times 10^{-8}$
10%	1.373 ± 0.025	$6.11 \times 10^{-8} \pm 1.77 \times 10^{-8}$
20%	1.941 ± 0.114	$1.24 \times 10^{-7} \pm 1.86 \times 10^{-8}$

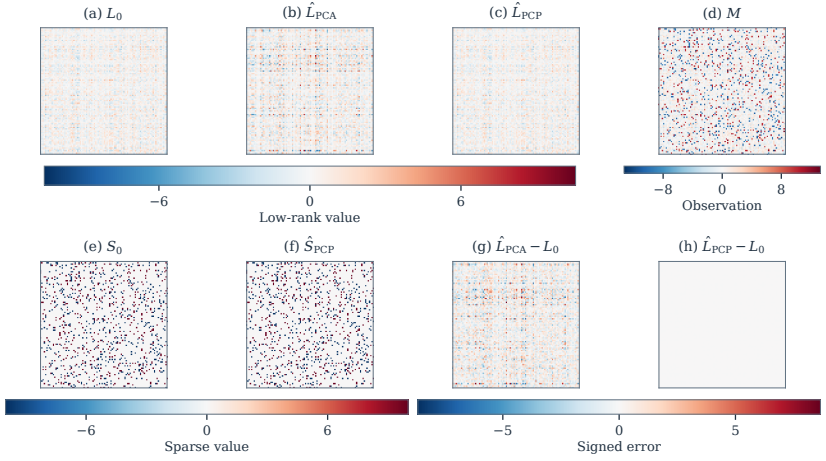
Relative Frobenius error, mean $\pm$ sample SD

Corruption energy

$$\frac{\|S_0\|_{\mathrm{F}}}{\sqrt{\mathbb{E}\,\|L_0\|_{\mathrm{F}}^2}} = 10\sqrt{p}.$$

- $1.41\times$ the signal scale at just 2%
- SVD directions pulled toward large errors in M
- PCP allocation of isolated errors to S

Component recovery at 10% corruption



- Trial 1: $E_L = 1.367$ for SVD, $E_L = 8.27 \times 10^{-8}$ for PCP, fitted PCP rank 5
- Shared scales for panels (a)–(c), (e)–(f), and (g)–(h)

Corruption support recovery



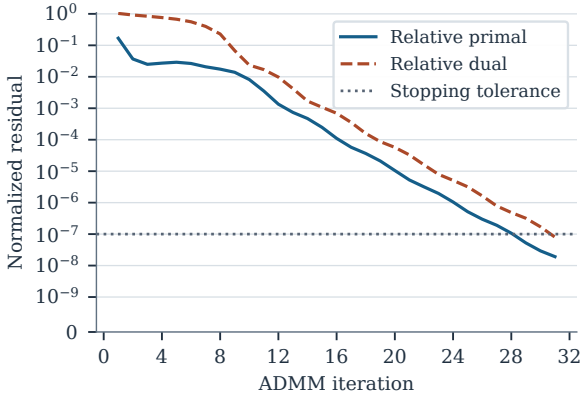
10% corruption, trial 1

- 1000 detections, 0 false positives, 0 misses
- Threshold $|\hat{S}_{ij}| > 1$ for errors of magnitude 10

Across the corrupted main trials

- 18500 true corruptions, 0 false positives, 0 misses
- Mean F1 of 1.000 at each nonzero density

Convergence and computational cost



10% corruption, trial 1, 31 iterations

$$\rho_{\mathrm{p}}^{k+1} = \frac{\left\| M - L^{k+1} - S^{k+1} \right\|_{\mathrm{F}}}{\|M\|_{\mathrm{F}}},$$

$$\rho_{\mathrm{d}}^{k+1} = \frac{\mu \left\| S^{k+1} - S^{k} \right\|_{\mathrm{F}}}{\max(1, \|\mu Y^{k+1}\|_{\mathrm{F}})}.$$

- Feasibility and stationarity residuals both $\leq 1 \times 10^{-7}$
- 25/25 main runs converged, 2–40 iterations
- Mean time: 1.34 ms (SVD), 41.34 ms (PCP)
- One economy SVD per PCP iteration

Dense Gaussian noise extension

$$M = L_0 + S_0 + \sigma Z, \qquad Z_{ij} \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad p = 10\%.$$

Means across 5 trials per level

σ	SVD E_L	PCP E_L	PCP rank	F1
0.0	1.318	5.54×10^{-8}	5.0	1.000
0.1	1.320	0.075	54.8	1.000
0.5	1.345	0.359	55.0	0.914

Shared L_0, S_0, Z within each trial across all σ

Effect of model mismatch

- Dense noise absorbed into the fitted components under exact equality
- 946 false positives at $\sigma = 0.5$
- All 15/15 fits numerically converged

- Stable PCP [6]: residual budget $\|M - L - S\|_{\mathrm{F}} \leq \varepsilon$ for dense noise

Conclusions and scope

Evidence from the experiment

- H1 supported across the tested sparse corruption sweep
- Near-exact PCP signal recovery with accurate support detection
- SVD as the faster fit in the clean, known-rank control
- Repeated decompositions as the main PCP cost

Limits of the evidence

- Synthetic Gaussian factors and one gross error magnitude
- Descriptive comparisons from five trials per density
- Coherent signal or clustered errors as sources of ambiguity
- Dense noise as a source of higher fitted rank and false alarms

References

[1] Eckart and Young, *The Approximation of One Matrix by Another of Lower Rank*, 1936

[2] Jolliffe, *Principal Component Analysis*, 2nd ed., 2002

[3] Candès, Li, Ma, and Wright, *Robust Principal Component Analysis?*, 2011

[4] Cai, Candès, and Shen, *A Singular Value Thresholding Algorithm for Matrix Completion*, 2010

[5] Boyd et al., *Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers*, 2011

[6] Zhou et al., *Stable Principal Component Pursuit*, 2010